Echo-Code: Multi-Task Deep Learning for Automated Cardiac Geometric Feature Extraction from Ultrasound Echoes

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Abstract

Portable ultrasound devices have made cardiac imaging widely accessible, but interpreting echocardiograms still requires expert training. We present Echo-Code, a multi-task deep learning pipeline that produces a hemodynamic report from a single echocardiographic video by integrating three models: a ConvNeXt-Transformer ejection fraction (EF) regression model, an attention-gated U-Net for four-chamber segmentation, and an EfficientNet-B0 quality grader. We train across five public datasets totaling over 65,000 samples, handling the challenge that no single source provides all required labels. The segmentation model achieves source-domain Dice of 0.860 and target-domain Dice of 0.8090, EF regression reaches RMSE 5.38 and R^2 0.809, and quality classification accuracy is 97.3%. From segmentation masks we extract chamber areas, volumes, and thickness ratios as quantitative proxies for tissue-level structural organization relevant to cardiac remodeling assessment.

1 Introduction

Cardiac tissue remodeling, the structural reorganization of heart muscle in response to disease or injury, alters chamber geometry, wall thickness, and chamber proportions. These geometric changes reflect shifts in cellular architecture and extracellular matrix composition. From a materials science perspective, they are measurable proxies for structure-property relationships governing cardiac tissue function, relevant for tracking disease progression, evaluating regenerative therapies, and connecting tissue organization to organ performance. In tissue engineering contexts, for instance, monitoring how chamber geometry evolves after implantation of a cardiac patch requires exactly this kind of reproducible geometric quantification.

Echocardiography is the standard modality for cardiac structure evaluation, and portable devices now enable acquisition beyond hospitals. But interpreting these images still requires expertise. We present Echo-Code, a pipeline that takes a raw echocardiographic video and outputs a structured hemodynamic report by integrating EF regression, four-chamber segmentation, and quality grading. The system trains across five public datasets providing EF labels, partial and full chamber annotations, and quality grades. Our contributions: (1) a unified pipeline producing multiple cardiac measurements from a single video, (2) a multi-source segmentation strategy handling heterogeneous partial labels through an ignore-index mechanism, and (3) an analysis of cross-domain generalization under label scarcity.

2 Related Work

Ouyang et al. [1] introduced EchoNet-Dynamic (10,030 videos with expert EF labels), showing automated EF estimation can approach inter-observer variability. We extend their approach to include segmentation and quality assessment. For segmentation, U-Net [2] dominates medical imaging through its encoder-decoder structure with skip connections that recover spatial detail at multiple scales. Oktay et al. [3] added attention gates that suppress irrelevant features in skip connections, improving small-structure segmentation. CAMUS [4] and CardiacUDA [5] provide partial and full cardiac annotations respectively, but no single public dataset labels all four chambers across diverse domains, necessitating multi-source training. CACTUS [6] provides expert quality grades (0–10), enabling automated screening of unreliable acquisitions.

3 Methods

3.1 Data

We use five public datasets (Table 1). **EchoNet-Dynamic** [1] provides 10,033 apical four-chamber videos with expert EF values. **CAMUS** [4] contains 7,500 frames annotated for left ventricle (LV) and left atrium (LA) only. **CardiacUDA** [5] provides 9,680 frames with four-chamber labels across domains G (source) and R (held-out target). **PLOSONE** is a 202-frame subset of EchoNet-Dynamic manually segmented for all four chambers. **CACTUS** [6] supplies 37,757 images with expert quality grades (0–10). Segmentation label scarcity is the central constraint: CardiacUDA-G, our main four-chamber source, is small. We oversample it 4× and combine it with CAMUS and PLOSONE using an ignore-index mechanism where pixels lacking annotation for a given class receive label 255 and are excluded from the loss.

Table 1: Datasets, sample counts, and roles in Echo-Code.

Dataset	Samples	Labels	Pipeline Role
EchoNet-Dynamic [1]	10,033	EF, LV trace	EF regression (train + eval)
CAMUS [4]	7,500	LV, LA	Segmentation (left-side only)
CardiacUDA [5]	9,680	All 4 Chambers	Segmentation (G = source, R = held-out target)
PLOSONE	202	All 4 Chambers	Small fully-labeled supplement (subset of EchoNet-Dynamic)
CACTUS [6]	37,757	Grade 0–10	Image quality grading

3.2 Ejection Fraction Regression

The EF model takes a 32-frame clip and predicts a scalar EF. The image encoder is ConvNeXt-tiny [7], a modernized convolutional network using large 7×7 depthwise kernels and LayerNorm (which normalizes each sample's features independently, making training less sensitive to batch composition) instead of batch normalization. ConvNeXt achieves stronger ImageNet accuracy than ResNets at comparable size, producing richer per-frame features. Each frame yields a feature vector, and the 32-vector sequence receives sinusoidal positional encodings (fixed sine/cosine signals encoding frame position) before entering a 4-layer, 8-head Transformer encoder. The Transformer uses self-attention, where every frame computes weighted relationships to every other frame, capturing long-range temporal patterns across the cardiac cycle without the sequential bottleneck of recurrent networks or the limited temporal reach of 3D convolutions.

The Transformer output is mean-pooled and fed to a regression head with residual connections (shortcuts letting gradients bypass layers to prevent signal loss), GELU activations (a smooth activation that avoids ReLU's abrupt zeroing of negative values, improving gradient flow for regression), and a linear shortcut to the output. We initialize ConvNeXt from a pretrained cardiac model and apply differential learning rates: 5×10^{-6} for the backbone (preserving learned features) and 5×10^{-4} for the fresh head. The optimizer is AdamW, which adapts per-parameter learning rates using gradient statistics and applies weight decay (a penalty on large weights for regularization) directly. We use a weighted Huber loss, which behaves quadratically for small errors (smooth gradients) and linearly for large ones (avoiding outlier sensitivity), upweighting low-EF samples (2× for EF < 40, 3.5× for EF < 30) where clinical detection matters most. The learning rate follows cosine annealing (gradually reducing along a cosine curve to settle into a good minimum) with 5-epoch linear warmup. At validation, we average over 4 clips per video.

3.3 Four-Chamber Segmentation

The segmentation model is a U-Net [2], an encoder-decoder where the encoder compresses the image through pooling to capture what structures are present, the decoder upsamples to recover where they are, and skip connections pass fine-grained spatial details from each encoder level to the corresponding decoder level. Without skip connections, the decoder would reconstruct boundaries from the compressed bottleneck alone, losing precision on thin structures like chamber walls. We use four encoder levels (base 64 channels, doubling per level) producing a 24×24 bottleneck from 384×384 input.

Three choices target domain generalization. First, instance normalization replaces batch normalization. Batch normalization computes statistics across the training batch, memorizing source-domain intensity distributions that transfer poorly. Instance normalization computes statistics per-sample, stripping machine-specific intensity so the model learns structure rather than domain appearance. Second, attention gates [3] in the decoder learn spatial relevance maps: a gating signal from the decoder modulates encoder skip connections, suppressing noise from irrelevant regions and reducing oversegmentation. Third, bilinear upsampling replaces transposed convolutions, which produce checkerboard artifacts where output kernels overlap.

Deep supervision places auxiliary heads at two intermediate decoder levels (loss weights 0.4, 0.2), providing a direct gradient signal that prevents attenuation through four decoder stages. The loss combines cross-entropy (penalizing incorrect per-pixel class probabilities), Dice loss (directly optimizing predicted-true mask overlap, critical when classes occupy small image regions), and a boundary-aware term. RV receives 1.2× class weight for its smaller area. Training uses AdamW with cosine annealing warm restarts (periodically resetting the learning rate to escape local minima), gradient clipping at 5.0, and early stopping (patience 15). Test-time augmentation averages predictions over original and horizontally-flipped input.

3.4 Image Quality Grading

The quality model uses EfficientNet-B0 [8], which achieves a strong accuracy-to-size ratio through compound scaling (jointly optimizing depth, width, and resolution rather than one dimension alone). Two heads share the backbone: regression for 0–10 grade and 3-class classification (Bad: 0–3, Okay: 4–6, Good: 7–10). The loss sums Huber on the grade with weighted cross-entropy on the bucket, using inverse-frequency class weights for CACTUS's imbalanced distribution. Training: 30 epochs, AdamW, cosine annealing. The dual-head design lets both tasks share learned features about image sharpness, acoustic shadowing, and structural visibility, while the regression head provides a continuous grade that the classification head alone would discard by binning

3.5 Pipeline Integration

At inference, the quality grader flags unreliable frames. The segmentation model processes all frames; LV area over time identifies end-diastolic (ED, max area) and end-systolic (ES, min area) frames. Chamber areas from ED/ES masks convert to estimated volumes. The EF model provides a direct regression estimate. The report includes regression and segmentation-derived EF, EDV, ESV, stroke volume, heart rate, cardiac output, and quality grade.

4 Results and Discussion

4.1 Segmentation

The model achieves mean Dice of 0.860 on source validation and 0.784 on unseen target (CardiacUDA-R). Table 2 shows per-class scores. Left-side chambers (LV: 0.907, LA: 0.871) outperform right-side (RA: 0.735, RV: 0.713) on source, reflecting training composition: CAMUS contributes 7,500 LV/LA frames while four-chamber labels come only from smaller CardiacUDA-G and 202-frame PLOSONE.

Table 2: Per-class Dice on source validation and unseen target domain.

	LV	**LA**	**RA**	**RV**
Source (Val)	0.907	0.871	0.735	0.713
Target (R)	0.770	0.761	0.824	0.782

A notable cross-domain pattern: while overall target Dice drops 0.076, right-side chambers (RA: 0.824, RV: 0.782) *improve* on target relative to source. CardiacUDA-R likely has clearer right-chamber boundaries, and instance normalization helps adapt to target intensity. Left-side chambers degrade as expected (LV: −0.137, LA: −0.110), consistent with overfitting to abundant left-side annotations.

Figure 1: Segmentation predictions on target domain

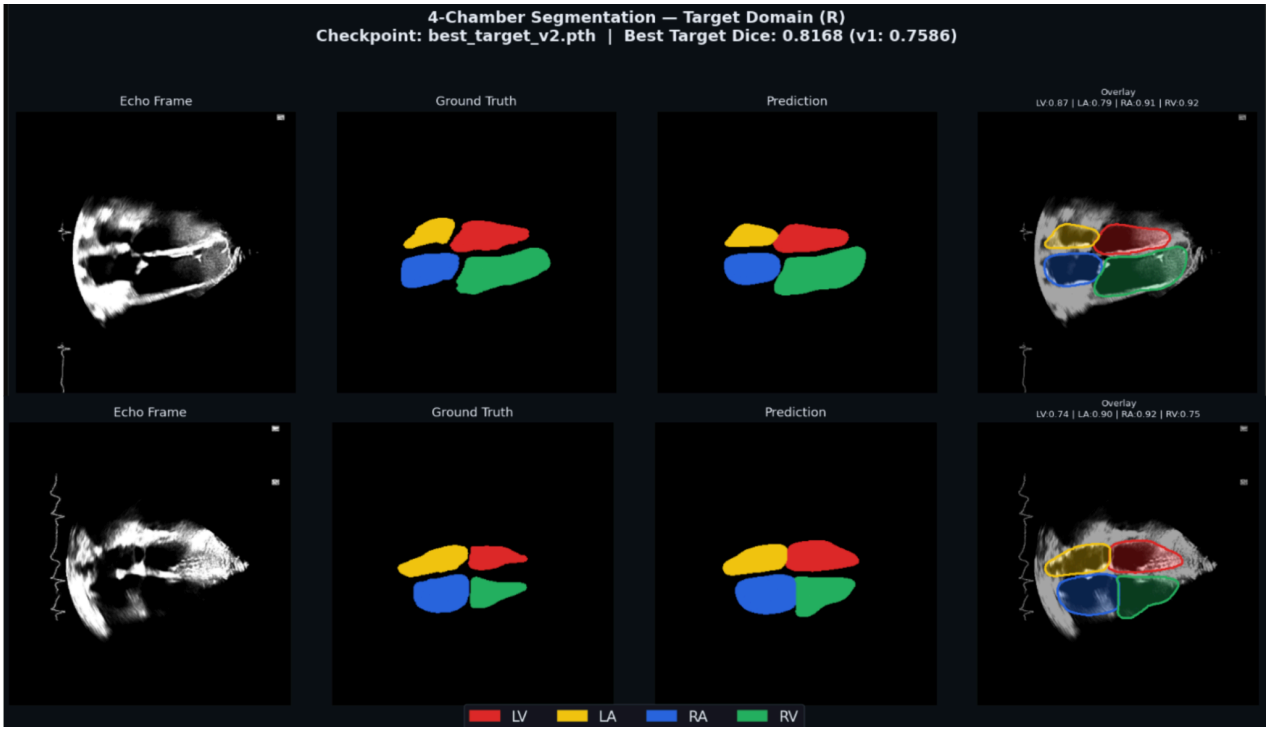
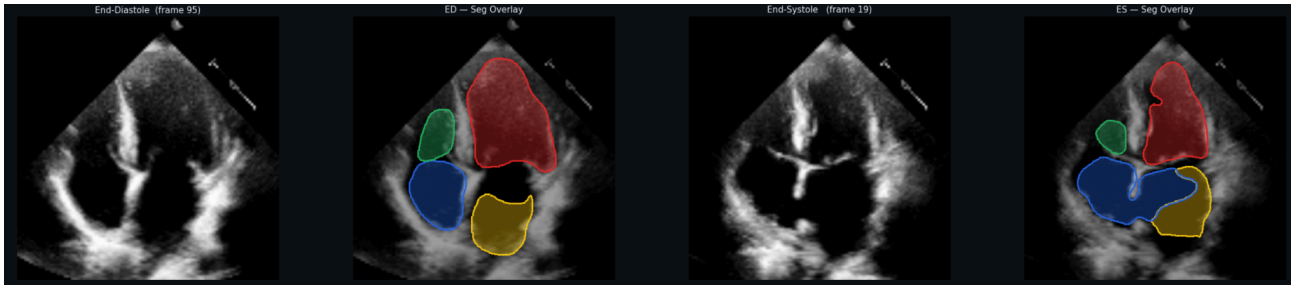


Figure 2: Cross model segmentation



4.2 Ejection Fraction

EF regression achieves an RMSE of 5.38, MAE of 4.00, and R^2 of 0.809 on the EchoNet-Dynamic validation set (1,288 videos). Inter-observer variability is typically 5–10 points, placing our model near the upper bound. Regression and segmentation-derived EF sometimes disagree, reflecting volume estimation calibration and segmentation error propagation.

4.3 Quality and Pipeline Output

Quality classification reaches 97.3% accuracy on the three-class CACTUS validation set (7,548 images), with a grade MAE of 0.13. The full pipeline produces hemodynamic reports with EF (both methods), EDV, ESV, stroke volume, heart rate, cardiac output, quality grade, and chamber areas at ED/ES. From a materials science perspective, these geometric measurements proxy for tissue-level organization: wall thickness changes, chamber dilation, and relative chamber sizes reflect extracellular matrix remodeling and myocyte hypertrophy, enabling longitudinal tracking relevant to tissue engineering and regenerative medicine.

Figure 3: Hemodynamic report and quality grading

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── Ejection Fraction ──────────────────────────
True EF              : 55.7 %
EF (regression)      : 61.2 %   ← primary
EF (seg-based)       : 50.7 %
Absolute error       : 5.6 pp

── Volumes ────────────────────────────────────
EDV                  : 58.2 mL   (true 60.5)
ESV                  : 28.7 mL   (true 26.8)
Stroke Volume        : 29.5 mL
ED frame             : 95
ES frame             : 19

── Cardiac Output ─────────────────────────────
Heart Rate (est.)    : 98 bpm
Cardiac Output       : 2.90 L/min

── Image Quality ──────────────────────────────
Label                : Bad
Grade                : 1.0/10
Confidence           : 100%
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4.4 Limitations

The 0.076 Dice domain gap means geometric features extracted from unseen domains carry meaningfully higher uncertainty, particularly for left-side structures where degradation is largest. Label scarcity is the core bottleneck: the entire four-chamber training signal comes from CardiacUDA-G (oversampled 4×) and the 202-frame PLOSONE set, which together provide far fewer annotated samples than robust multi-domain training requires. EF RMSE of ~9 points, at the upper end of inter-observer variability, limits fine-grained stratification between mild and moderate dysfunction. Volume estimation uses an area-based approximation rather than a biplane or model-of-revolution method, introducing systematic bias especially for asymmetric chambers. The pipeline assumes apical four-chamber input and does not perform view classification, restricting its applicability to pre-screened acquisitions.

5 Conclusion

Echo-Code integrates EF regression, four-chamber segmentation, and quality grading to produce automated hemodynamic reports from echocardiographic video. Trained across five heterogeneous datasets, the system demonstrates that practical cardiac analysis can be assembled from partially-annotated sources, though cross-domain generalization remains the primary challenge. The geometric features it extracts provide reproducible, quantitative proxies for cardiac structural organization that could support standardized assessment in both clinical monitoring and tissue engineering evaluation. Future directions include self-supervised pretraining to reduce annotation dependence, explicit domain adaptation to close the cross-domain gap, and multi-view fusion for more complete structural characterization connecting imaging features to tissue-level material properties.

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